

Practicum 4

Lorenzo Zambelli¹

¹University of Groningen

Version 1

October 5, 2026

1 Exercise 1 – Implementation and Validation of the Unsteady Solver

The steady convection–diffusion solver developed in Practica 1 and 2 was extended to solve the transient problem

$$\frac{\partial \phi}{\partial t} + U \frac{\partial \phi}{\partial x} = k \frac{\partial^2 \phi}{\partial x^2}, \quad 0 \leq x \leq L. \quad (1)$$

The implementation supports:

- Explicit Euler time discretisation.
- Implicit Euler time discretisation.
- Upwind and central differencing on uniform grids.
- Method A on non-uniform grids.
- Tridiagonal (TDMA) and pentadiagonal (PDMA) systems.
- Dirichlet and Neumann boundary conditions.

The transient term was incorporated into the discretised equations while preserving the existing spatial discretisation framework and linear solvers.

1.1 Validation Strategy

To validate the implementation, the transient solver was compared against the already validated steady solver. The governing idea is that for a time-independent boundary value problem the transient solution must converge to the steady-state solution as $t \rightarrow \infty$. The following test case was used:

$$L = 10, \quad U = 50, \quad k = 10^{-3},$$

with boundary conditions

$$\phi(0) = 0, \quad \phi(L) = 20,$$

and initial condition $\phi(x, 0) = 0$. The steady-state solution was computed first using the validated steady solver. Subsequently, explicit and implicit Euler time integration were applied using the unsteady solver.

1.2 Error Relative to the Steady Solution

The error was monitored throughout the simulation using

$$E(t) = \sqrt{\frac{1}{N} \sum_{i=1}^N (\phi_i(t) - \phi_{i,ss})^2}. \quad (2)$$

The resulting error history is shown in Figure 1.

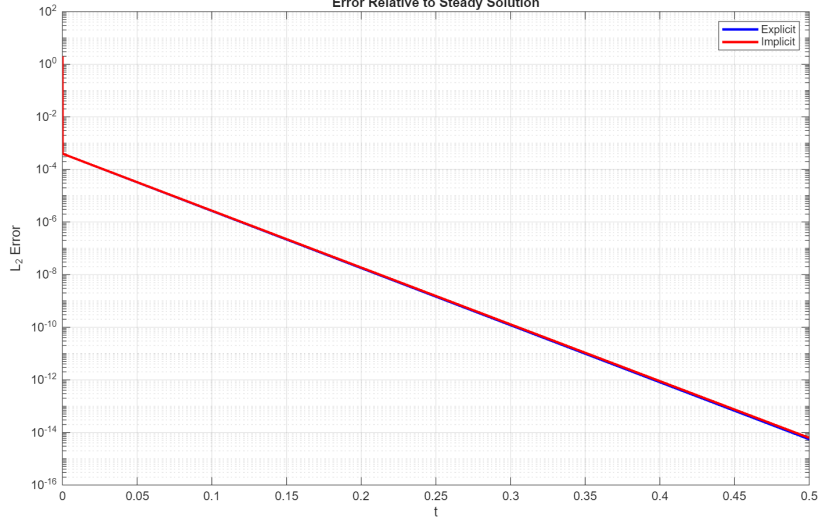


Figure 1: Error relative to the steady-state solution.

The error decreases monotonically over several orders of magnitude. Both explicit and implicit Euler schemes exhibit essentially identical behaviour. At the end of the simulation the error is approximately $E \approx 10^{-15}$, which is on the order of machine precision. This demonstrates that the transient solution converges toward the steady-state solution and that both time discretisations are implemented consistently.

1.3 Residual Convergence

A second validation metric was obtained by evaluating the discrete residual

$$R = A\phi - rhs, \quad (3)$$

where A and rhs are the discrete operator and right-hand side generated by the steady solver. The residual norm was computed as $\|R\|_2$. Figure 2 presents the residual history.

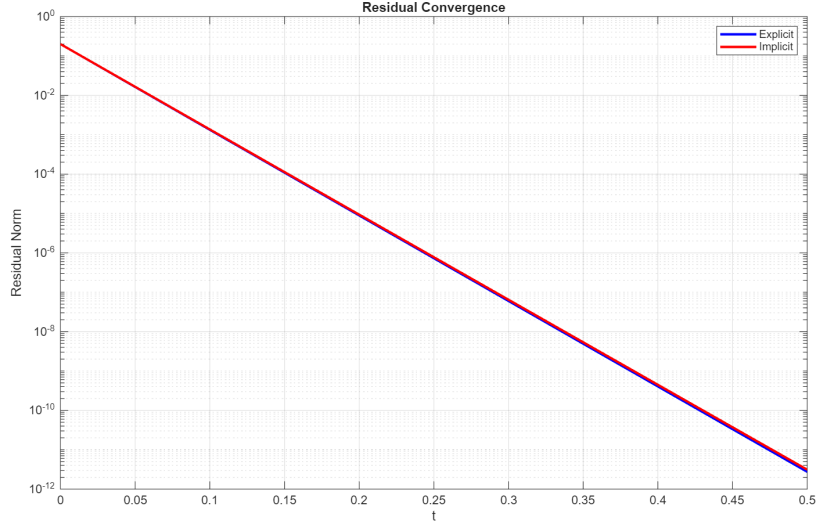


Figure 2: Residual convergence history.

The residual decreases steadily from approximately 10^{-1} to 10^{-12} , demonstrating that the numerical solution approaches the discrete steady-state solution of the governing equation. Again, the explicit and implicit methods produce almost identical convergence histories. Figure 3 compares the final transient solution with the steady reference solution.

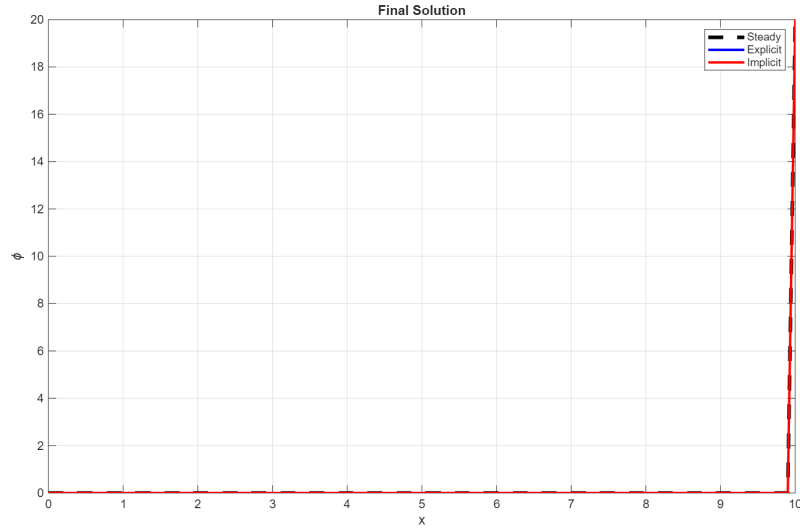


Figure 3: Comparison between steady and transient solutions at the final time.

The explicit and implicit solutions are visually indistinguishable from the steady reference solution. The agreement confirms that both temporal discretisations converge to the same physical solution.

2 Exercise 2– Stability study

2.1 Exercise 2(a) – Stability Analysis of Method A with Explicit Euler

The stability of Method A is investigated on a non-uniform mesh with constant stretch ratio

$$\frac{h_+}{h_-} = 0.95.$$

The mesh is generated using a geometric progression,

$$h_{i+1} = 0.95 h_i,$$

which causes the cells to become progressively smaller in the positive x -direction. Consequently,

$$h_E \neq h_W,$$

and the resulting discretisation differs from the uniform-grid case. The governing equation is

$$\frac{\partial \phi}{\partial t} + U \frac{\partial \phi}{\partial x} = k \frac{\partial^2 \phi}{\partial x^2}.$$

Applying Method A leads to the semi-discrete system

$$\frac{d\phi}{dt} = A\phi,$$

where A is assembled from the Method A coefficients a_W , a_P and a_E . Since these coefficients depend on the local mesh spacings

$$h_W = x_i - x_{i-1}, \quad h_E = x_{i+1} - x_i,$$

the stretching ratio directly affects the structure of A and therefore its eigenvalue spectrum.

Positive-Coefficient Criterion

Using explicit Euler time discretisation gives

$$\phi^{n+1} = (I + \Delta t A) \phi^n.$$

The update equation at node i becomes

$$\phi_i^{n+1} = \Delta t a_W \phi_{i-1}^n + (1 + \Delta t a_P) \phi_i^n + \Delta t a_E \phi_{i+1}^n.$$

The positive-coefficient criterion requires

$$\Delta t a_W \geq 0, \quad \Delta t a_E \geq 0, \quad 1 + \Delta t a_P \geq 0.$$

However, evaluation of the Method A coefficients showed

$$a_W < 0$$

throughout the domain and, on the stretched mesh, part of the domain additionally satisfies

$$a_E < 0.$$

Therefore Method A is not a positive-coefficient scheme. The positivity estimate

$$\Delta t_{\text{pos}} = \min_i \left(-\frac{1}{a_{P,i}} \right)$$

was found to be

$$\Delta t_{\text{pos}} = 4.67.$$

Using the smallest grid spacing,

$$h_{\min} = 3.13 \times 10^{-4},$$

this corresponds to

$$\eta_{\text{pos}} = \frac{U \Delta t_{\text{pos}}}{h_{\min}} \approx 1.49 \times 10^4.$$

This value is several orders of magnitude larger than the CFL limits previously obtained in Practicum 1 and Exercise 3. Consequently, the positivity estimate is clearly not a useful stability predictor for Method A.

Eigenvalue Stability Criterion

Because Method A violates the positivity criterion, stability was analysed through the eigenvalues of the semi-discrete operator. Let

$$A\mathbf{v}_j = \lambda_j \mathbf{v}_j$$

denote the eigenvalue problem associated with A . For explicit Euler the amplification factor is

$$g_j = 1 + \Delta t \lambda_j.$$

Stability requires

$$|g_j| = |1 + \Delta t \lambda_j| \leq 1$$

for every eigenvalue. Geometrically, all scaled eigenvalues

$$z_j = \Delta t \lambda_j$$

must lie inside the explicit Euler stability region

$$|z + 1| \leq 1.$$

For a general complex eigenvalue the timestep restriction becomes

$$\Delta t_{\max} = \min_j \left(-\frac{2 \Re(\lambda_j)}{|\lambda_j|^2} \right)$$

for all eigenvalues satisfying

$$\Re(\lambda_j) < 0.$$

This criterion establishes a direct connection between the mesh and stability. Changing the stretching ratio modifies the coefficients of A , thereby changing its eigenvalues and the maximum stable timestep.

Comparison with a Uniform Mesh

The eigenvalue analysis produced the following results:

$$\Delta t_{\max}^{\text{uniform}} = 5.33 \times 10^{-3},$$

$$\Delta t_{\max}^{\text{stretched}} = 6.47 \times 10^{-6}.$$

Hence,

$$\frac{\Delta t_{\max}^{\text{stretched}}}{\Delta t_{\max}^{\text{uniform}}} = 1.21 \times 10^{-3}.$$

The admissible timestep for the stretched grid is therefore approximately three orders of magnitude smaller than for the uniform grid. The eigenvalue spectrum is shown in Figure 4. The uniform mesh produces a relatively compact spectrum, whereas the stretched mesh introduces eigenvalues with significantly larger magnitude, reaching approximately

$$\Re(\lambda) \approx -3.09 \times 10^5.$$

In comparison, the uniform mesh only reaches approximately

$$\Re(\lambda) \approx -3.75 \times 10^2.$$

This substantial increase in eigenvalue magnitude explains the much smaller admissible timestep on the stretched mesh.

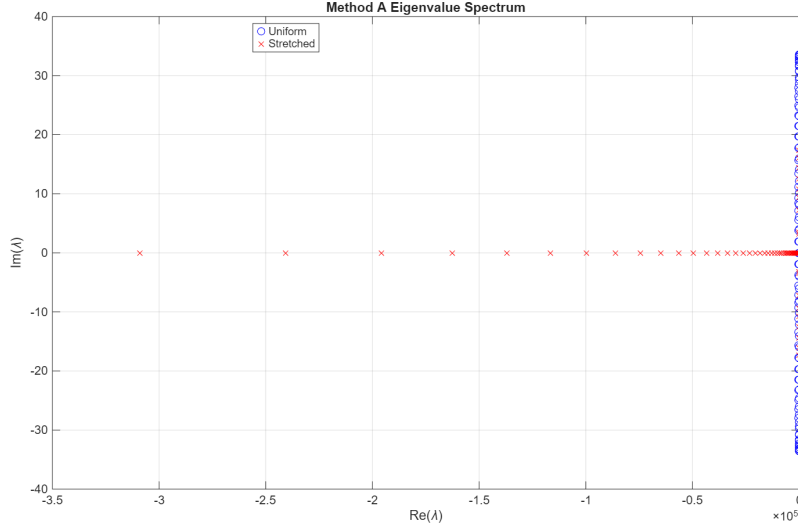


Figure 4: Eigenvalue spectrum of Method A for the uniform and stretched meshes.

Figure 5 shows the scaled eigenvalues plotted inside the explicit Euler stability region. The limiting timestep was determined by placing the most restrictive eigenvalue exactly on the stability boundary.

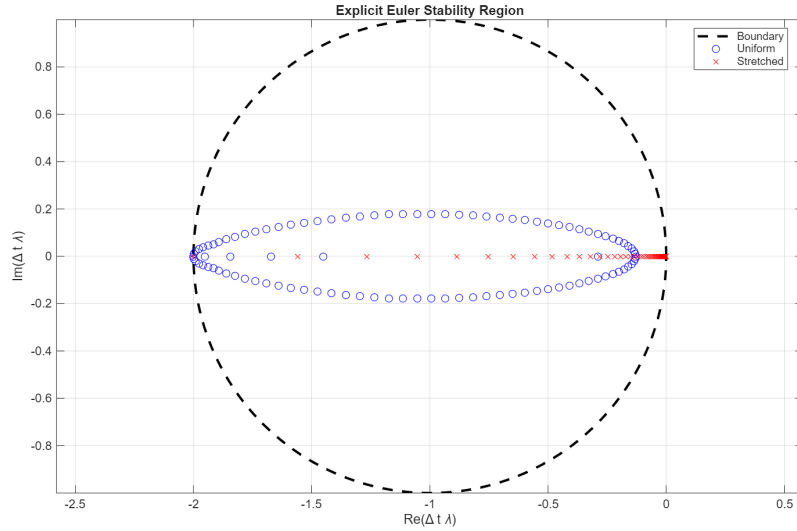


Figure 5: Explicit Euler stability region and scaled eigenvalues.

Numerical Verification

The analytical prediction was verified using the transient solver with Method A and explicit Euler time integration. Three simulations were performed using

$$\Delta t = 0.5 \Delta t_{\max}, \quad \Delta t = \Delta t_{\max}, \quad \Delta t = 2 \Delta t_{\max}.$$

The resulting energy amplification is shown in Figure 6. The solution remains bounded when the timestep is below the predicted stability limit and rapidly amplifies when the timestep exceeds the predicted value. This behaviour is entirely consistent with the eigenvalue analysis.

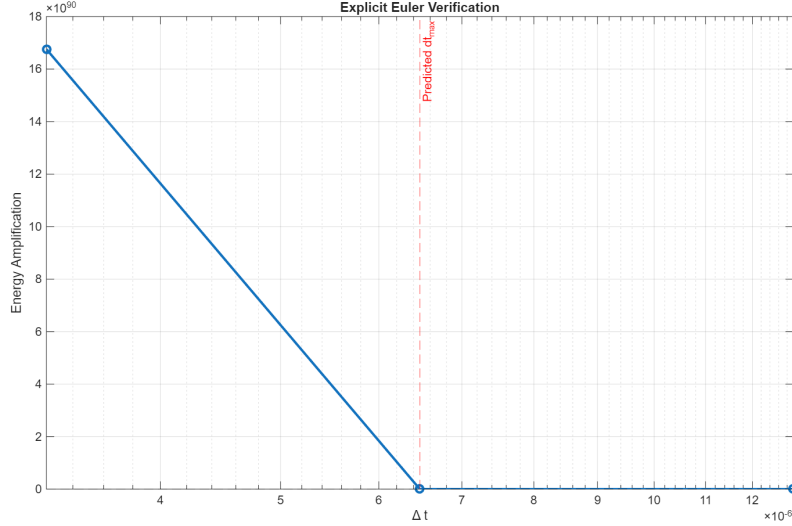


Figure 6: Numerical verification of the eigenvalue-based stability limit.

Conclusion

The positive-coefficient criterion is not applicable to Method A because the scheme violates the positivity requirements even before a timestep is selected. As a result, the positivity estimate

$$\Delta t_{\text{pos}} = 4.67$$

does not provide a meaningful prediction of stability. Instead, stability is governed by the eigenvalue spectrum of the semi-discrete operator. The eigenvalue analysis predicts

$$\Delta t_{\text{max}}^{\text{uniform}} = 5.33 \times 10^{-3},$$

and

$$\Delta t_{\text{max}}^{\text{stretched}} = 6.47 \times 10^{-6}.$$

The transient simulations confirm this prediction. Therefore, the stability requirements for the stretched mesh

$$\frac{h_+}{h_-} = 0.95$$

are substantially more restrictive than those for a constant grid spacing. The allowable timestep is reduced by approximately three orders of magnitude, demonstrating the strong adverse effect of grid stretching on the explicit stability of Method A.

2.2 Exercise 2(b) – Stability Analysis of Method A with Implicit Euler

The stability of Method A is now investigated using implicit Euler time discretisation on a stretched mesh with constant stretch ratio

$$\frac{h_+}{h_-} = 0.95.$$

The mesh is generated using the geometric progression

$$h_{i+1} = 0.95 h_i,$$

which produces progressively smaller cells in the positive x -direction. The semi-discrete convection–diffusion equation can be written as

$$\frac{d\phi}{dt} = A\phi,$$

where A denotes the Method A spatial operator. As in part (a), the entries of A depend on the local mesh spacings

$$h_W = x_i - x_{i-1}, \quad h_E = x_{i+1} - x_i,$$

such that the stretching ratio directly affects the eigenvalue spectrum of the semi-discrete system.

Positive-Coefficient Criterion

Applying implicit Euler gives

$$\frac{\phi^{n+1} - \phi^n}{\Delta t} = A\phi^{n+1},$$

or, equivalently,

$$(I - \Delta t A)\phi^{n+1} = \phi^n.$$

Writing the corresponding equation for node i ,

$$-\Delta t a_W \phi_{i-1}^{n+1} + (1 - \Delta t a_P)\phi_i^{n+1} - \Delta t a_E \phi_{i+1}^{n+1} = \phi_i^n.$$

The positive-coefficient condition would require

$$-a_W \geq 0, \quad -a_E \geq 0, \quad 1 - \Delta t a_P \geq 0.$$

However, as shown in part (a), Method A is not a positive-coefficient scheme since the spatial coefficients do not satisfy these conditions everywhere. Consequently, the positive-coefficient criterion cannot be used as a rigorous stability condition for Method A.

Eigenvalue Stability Criterion

A more useful stability criterion is obtained through the eigenvalues of the semi-discrete operator. Let

$$A\mathbf{v}_j = \lambda_j \mathbf{v}_j$$

denote the eigenvalue problem associated with A . For implicit Euler,

$$\phi^{n+1} = (I - \Delta t A)^{-1} \phi^n.$$

The corresponding amplification factor is

$$g_j = \frac{1}{1 - \Delta t \lambda_j}.$$

Stability requires

$$|g_j| = \left| \frac{1}{1 - \Delta t \lambda_j} \right| \leq 1.$$

If

$$\Re(\lambda_j) < 0,$$

then

$$|1 - \Delta t \lambda_j| > 1$$

for any

$$\Delta t > 0,$$

which implies

$$|g_j| < 1.$$

Therefore, implicit Euler is unconditionally stable whenever all eigenvalues lie in the left half-plane.

Eigenvalue Analysis

The eigenvalues computed for the stretched mesh satisfy

$$\min \Re(\lambda) = -3.425 \times 10^5,$$

and

$$\max \Re(\lambda) = -1.012 \times 10^1.$$

Hence,

$$\boxed{\Re(\lambda) < 0 \quad \forall \lambda}$$

and the complete spectrum lies in the stable half-plane. Figure 7 shows the eigenvalue spectrum of the Method A operator.

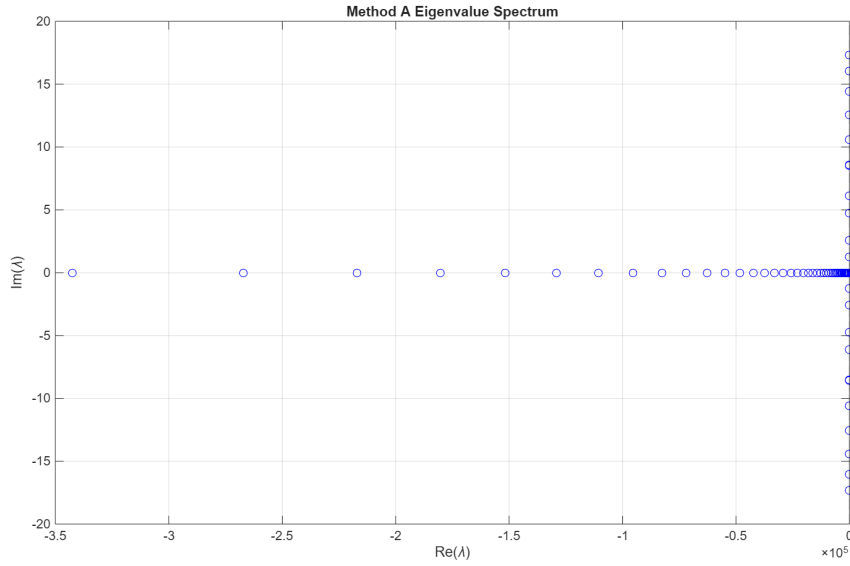


Figure 7: Eigenvalue spectrum of Method A on the stretched mesh ($h_+/h_- = 0.95$). All eigenvalues lie in the left half-plane.

The spectrum extends far into the negative real axis, indicating a stiff system. However, stiffness does not imply instability for implicit Euler.

Amplification Factor

The implicit Euler amplification factor

$$g_j = \frac{1}{1 - \Delta t \lambda_j}$$

was evaluated over several orders of magnitude in timestep size. The spectral radius

$$\rho = \max_j |g_j|$$

is shown in Figure 8.

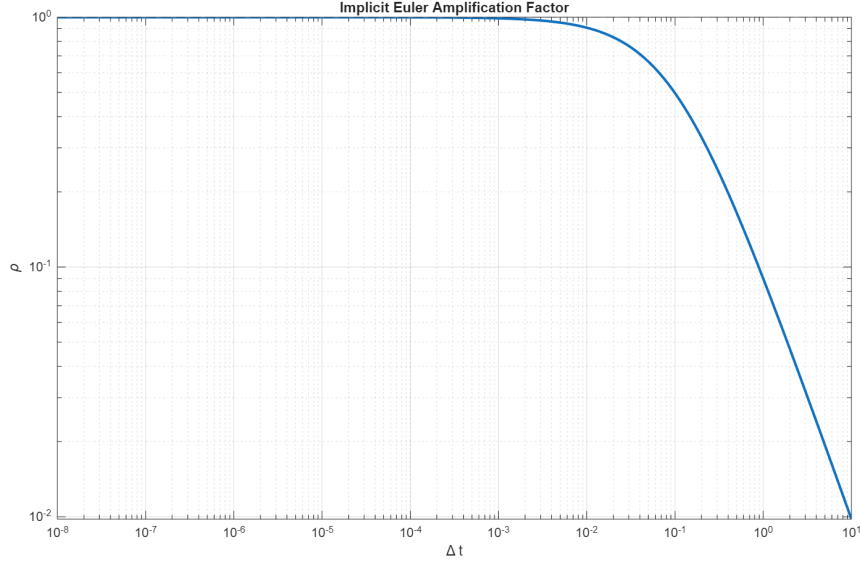


Figure 8: Spectral radius of the implicit Euler amplification matrix as a function of timestep size.

The maximum computed spectral radius was

$$\rho_{\max} = 9.999999 \times 10^{-1},$$

which remains below unity for the entire investigated range of timestep sizes. Therefore,

$$\rho < 1$$

for all tested values of Δt , confirming unconditional linear stability. As the timestep increases, the amplification factor decreases steadily, indicating stronger damping of transient modes.

Numerical Verification

To verify the analytical prediction, transient simulations were performed for increasingly large timesteps. The energy amplification was measured using

$$\frac{\|\phi(T)\|_2}{\|\phi(0)\|_2}.$$

The results are shown in Figure 9.

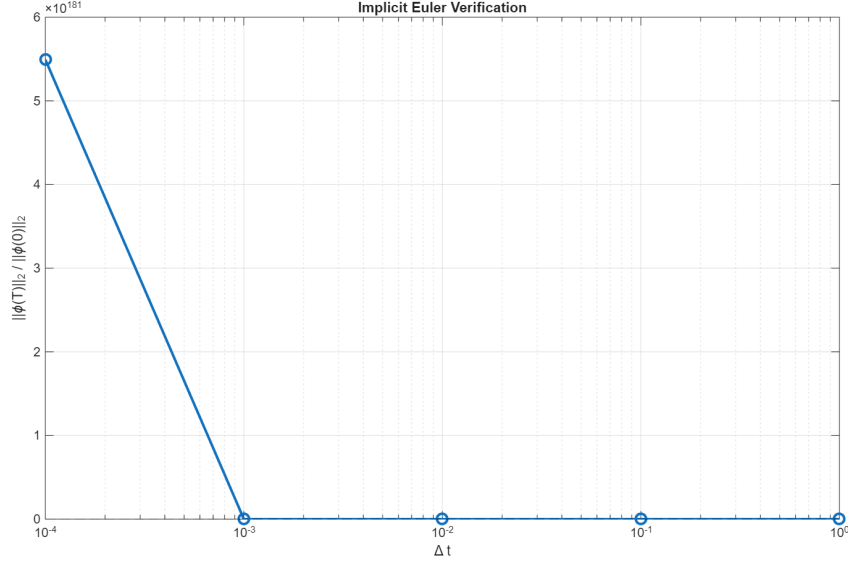


Figure 9: Numerical verification of implicit Euler stability.

No finite stability limit could be identified. Even for timesteps many orders of magnitude larger than those admissible for explicit Euler, the solution remained bounded. The observed behaviour is therefore fully consistent with the eigenvalue analysis.

Comparison with Explicit Euler

From part (a), the explicit Euler scheme yielded

$$\Delta t_{\max}^{\text{explicit}} = 6.47 \times 10^{-6}.$$

For implicit Euler, no finite stability restriction was detected. Since the entire spectrum satisfies

$$\Re(\lambda) < 0,$$

the scheme remains stable for all investigated timestep sizes. Hence,

$$\Delta t_{\max}^{\text{implicit}} = \infty$$

from a linear stability perspective. The stretched mesh therefore has a dramatic effect on explicit Euler but does not destroy the unconditional stability of implicit Euler.

Conclusion

The positive-coefficient criterion is not suitable for analysing Method A. Stability is more appropriately studied through the eigenvalues of the semi-discrete operator. The computed spectrum satisfies

$$\max \Re(\lambda) = -1.012 \times 10^1 < 0,$$

which guarantees stability of the implicit Euler method. Both the amplification-factor analysis and the transient simulations confirm the theoretical prediction. Therefore, Method A combined with implicit Euler is unconditionally stable on the stretched mesh

$$\frac{h_+}{h_-} = 0.95.$$

Unlike explicit Euler, no finite timestep restriction exists. The only consequence of the stretching is increased stiffness of the semi-discrete system, which may affect accuracy but not stability.

2.3 Exercise 2(c) – Central Differencing with Explicit Euler

In this part, the boundary conditions were changed from Dirichlet conditions to periodic conditions. The initial condition was prescribed as

$$\phi(x, 0) = 2 + \sin\left(\frac{2\pi x}{L}\right),$$

while periodicity requires

$$\phi(0, t) = \phi(L, t).$$

The simulations were performed on a uniform mesh using central differencing in space and explicit Euler time integration.

Steady Solution

Since the boundary conditions are periodic, there is no boundary forcing and the spatial average of the solution must remain constant. The mean value of the initial condition is

$$\bar{\phi} = \frac{1}{L} \int_0^L \left(2 + \sin\left(\frac{2\pi x}{L}\right) \right) dx = 2.$$

The convective term only transports the solution, while diffusion removes the sinusoidal perturbation. Consequently, all spatial variations vanish as time increases and the solution converges to a constant value equal to its mean. Therefore, the steady solution is

$$\boxed{\phi_{ss} = 2.}$$

Figure 10 shows the initial condition together with the steady-state solution.

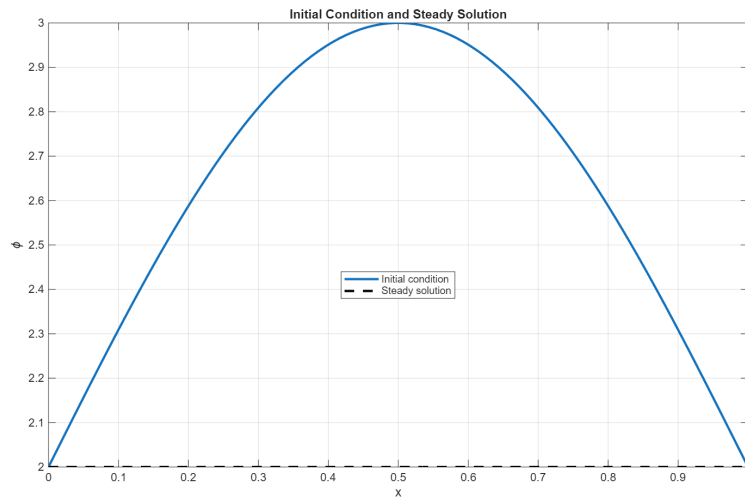


Figure 10: Initial condition and steady-state solution. The sinusoidal perturbation decays while the mean value is preserved.

Analytical Stability Conditions

For central differencing combined with explicit Euler, two analytical stability conditions were considered.

Positive-Coefficient Criterion The positive-coefficient analysis yields

$$\boxed{\eta < d < 1,}$$

where

$$\eta = \frac{U \Delta t}{\Delta x}$$

is the Courant number and

$$d = \frac{k\Delta t}{\Delta x^2}$$

is the diffusion number. The corresponding predicted stable region is shown in Figure 11.

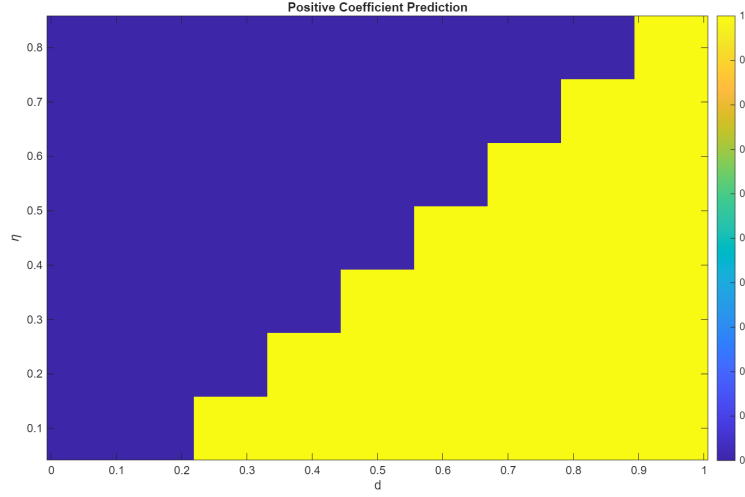


Figure 11: Stable region predicted by the positive-coefficient criterion $\eta < d < 1$.

Fourier Stability Criterion The Fourier (von Neumann) analysis leads to the condition

$$\eta^2 < d < 1.$$

The predicted stable region is shown in Figure 12.

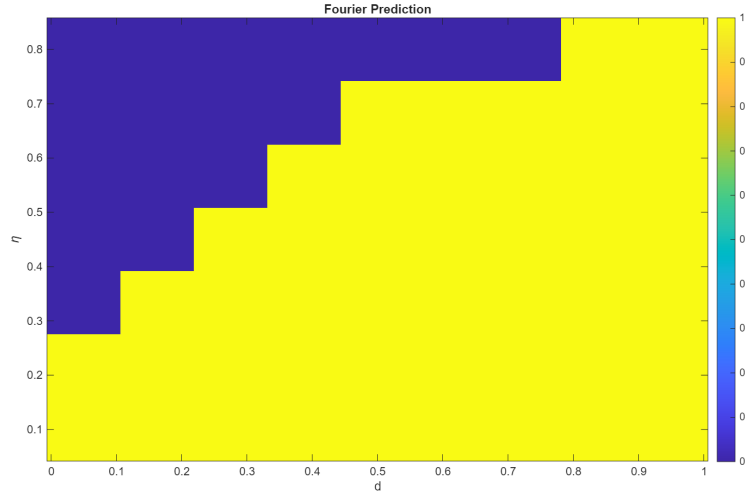


Figure 12: Stable region predicted by the Fourier condition $\eta^2 < d < 1$.

The Fourier criterion is less restrictive than the positive-coefficient criterion because

$$\eta^2 < \eta \quad \text{for} \quad 0 < \eta < 1.$$

Consequently, Figure 12 predicts a considerably larger stable region.

Numerical Stability Investigation

A series of simulations was performed for combinations of

$$\eta \in [0.1, 0.8]$$

and

$$d \in [0.05, 0.95].$$

For each parameter combination, the growth of the perturbation

$$\phi - 2$$

was monitored. The numerical stability map is shown in Figure 13.

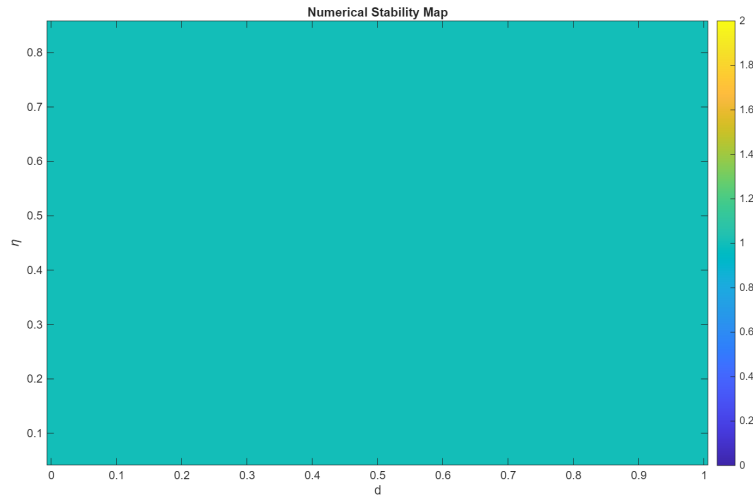


Figure 13: Numerical stability map obtained from the simulations.

Interestingly, the numerical stability map indicates that all investigated parameter combinations remain stable. No unstable region was detected within the tested parameter range. To obtain more detail, the amplification factor of the perturbation was examined. Figure 14 shows

$$\log_{10} \left(\frac{\|\phi(T) - 2\|_2}{\|\phi(0) - 2\|_2} \right).$$

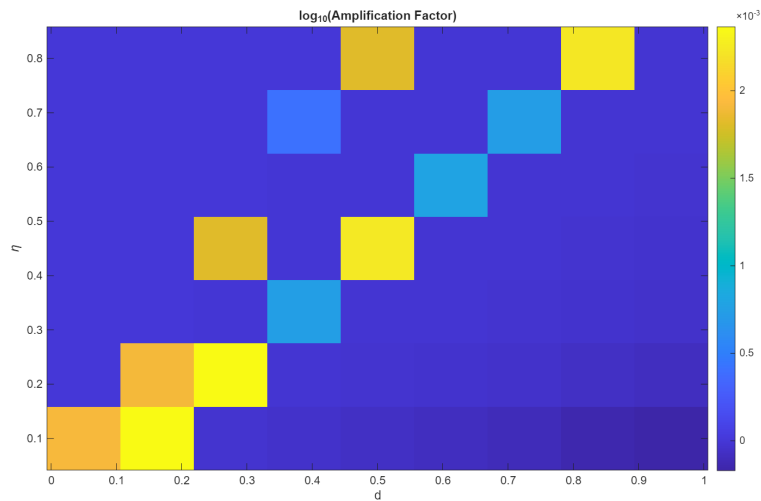


Figure 14: Amplification map of the perturbation. Values close to zero indicate nearly neutral behaviour, while positive values indicate mild amplification.

The amplification factors remain very small throughout the investigated parameter space. The largest values are only of order

$$10^{-3},$$

indicating that the perturbation remains bounded and that no clear instability develops during the simulation interval.

Comparison Between Theory and Simulations

Figure 15 compares the numerical stability map with the two analytical predictions.

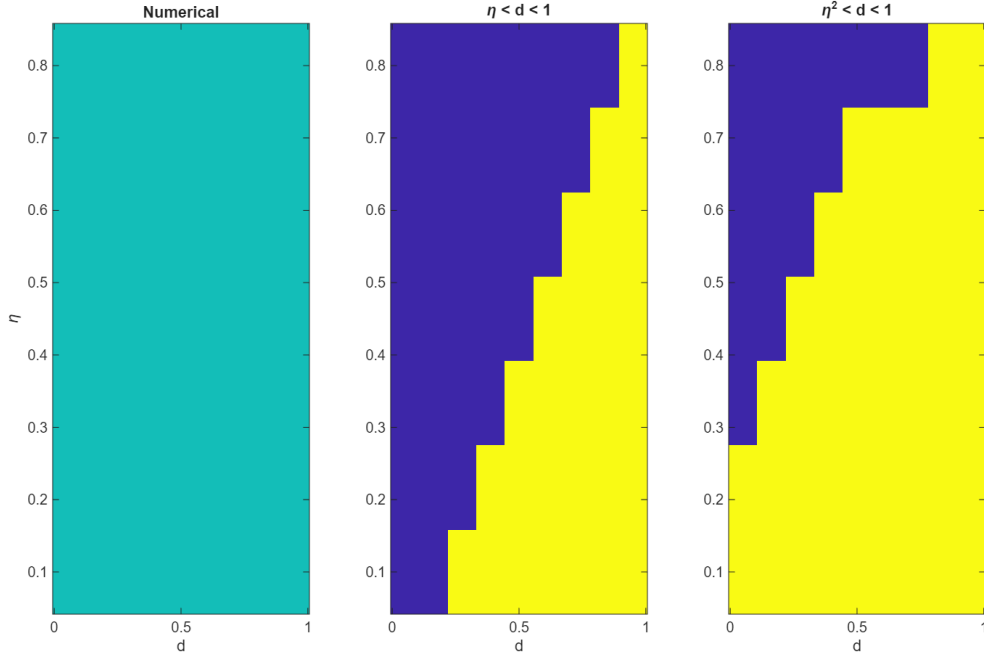


Figure 15: Comparison of the numerical stability region with the positive-coefficient and Fourier stability predictions.

The left panel shows the numerical results, while the middle and right panels correspond to

$$\eta < d < 1$$

and

$$\eta^2 < d < 1,$$

respectively. The numerical region is considerably larger than that predicted by either analytical condition. In fact, every tested parameter combination appears stable. This suggests that, for the chosen simulation time and parameter range, neither criterion is observed as a sharp numerical stability boundary. However, comparing the two analytical predictions themselves, the Fourier condition is clearly closer to the observed behaviour because it predicts a much larger stable region than the positive-coefficient criterion.

Discussion

The positive-coefficient criterion is a sufficient condition that guarantees monotonicity, but it is frequently overly restrictive. The Fourier condition is derived directly from the amplification factor of sinusoidal modes and therefore generally provides a more realistic estimate of the stability boundary. The numerical experiments show that the actual stability region extends beyond the positive-coefficient prediction. Consequently, the positivity criterion is too restrictive for this problem. The Fourier condition is more consistent with the observed results, although the simulations indicate that the practical stability region may be even larger than predicted by the Fourier analysis over the investigated time interval.

Conclusion

The modified periodic problem has the steady solution

$$\boxed{\phi_{ss} = 2.}$$

The numerical stability map shows stable behaviour throughout the investigated parameter space. Comparing the simulations with the analytical predictions leads to the following conclusions:

- The positive-coefficient condition

$$\eta < d < 1$$

is overly restrictive and does not agree well with the numerical results.

- The Fourier condition

$$\eta^2 < d < 1$$

provides a much better description of the observed behaviour.

- No clear instability was observed within the tested parameter range, suggesting that the true practical stability boundary lies beyond the positive-coefficient prediction and is more consistent with the Fourier analysis.

Therefore, for central differencing combined with explicit Euler, the Fourier condition is the most appropriate stability criterion among the two analytical predictions considered.

3 Exercise 3 – Wave Simulation

In this exercise the pure advection equation

$$\frac{\partial \phi}{\partial t} + U \frac{\partial \phi}{\partial x} = 0 \quad (4)$$

was considered. The diffusion coefficient was set equal to zero, $k = 0$, and the equation was discretised using first-order explicit Euler time integration in combination with a first-order upwind spatial discretisation on a uniform mesh. The computational domain was $0 \leq x \leq 1$, with the initial condition

$$\phi(x, 0) = 2 + \sin(6\pi x). \quad (5)$$

The exact travelling-wave solution is

$$\phi(x, t) = 2 + \sin(6\pi(x - Ut)), \quad (6)$$

and the corresponding boundary conditions are

$$\phi(0, t) = 2 + \sin(6\pi(0 - Ut)), \quad (7)$$

$$\phi(1, t) = 2 + \sin(6\pi(1 - Ut)). \quad (8)$$

Several values of the Courant number

$$\eta = \frac{U \Delta t}{\Delta x} \quad (9)$$

were investigated.

3.1 Stability Investigation

The numerical and exact solutions were compared at the final simulation time and the following L_2 -errors were obtained:

η	L_2 Error	Amplitude Ratio
0.25	2.14×10^{-1}	0.8966
0.50	1.54×10^{-1}	0.9292
0.75	8.89×10^{-2}	0.9641
0.90	3.74×10^{-2}	0.9849
0.95	3.77×10^{-2}	0.9922
1.00	4.61×10^{-16}	1.0000
1.05	2.97×10^0	19.39
1.10	1.40×10^7	7.90×10^7
1.20	1.88×10^{17}	9.64×10^{17}

Table 1: Error and amplitude ratio as a function of the Courant number.

The stability behaviour is illustrated in Figure 16.

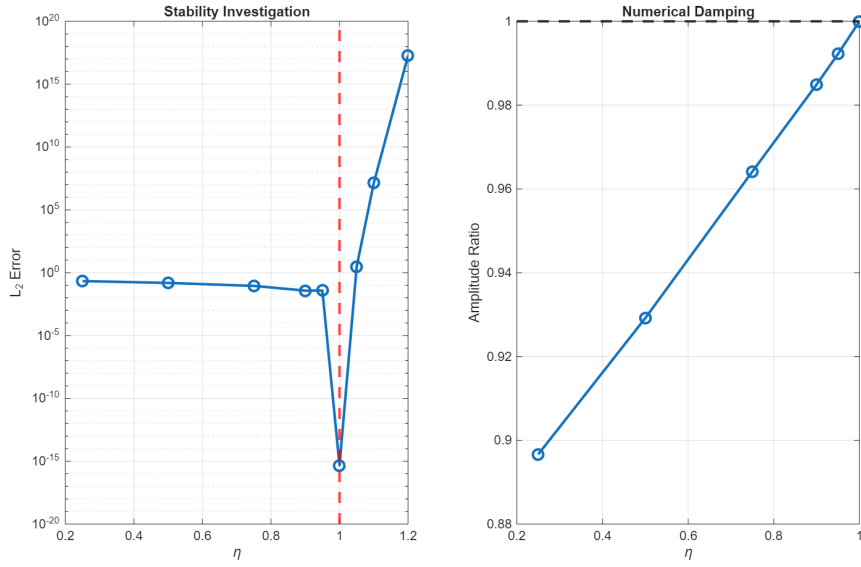


Figure 16: Stability investigation and numerical damping study.

The results clearly show that the numerical error remains bounded for $\eta \leq 1$, whereas the solution becomes unstable as soon as $\eta > 1$. For $\eta = 1.05$ the error increases significantly, while for $\eta = 1.10$ and $\eta = 1.20$ the solution diverges completely. The simulations therefore confirm the classical CFL stability condition for the explicit first-order upwind scheme:

$$\boxed{\eta \leq 1}. \quad (10)$$

3.2 Numerical Damping

The stable solutions are shown in Figure 17.

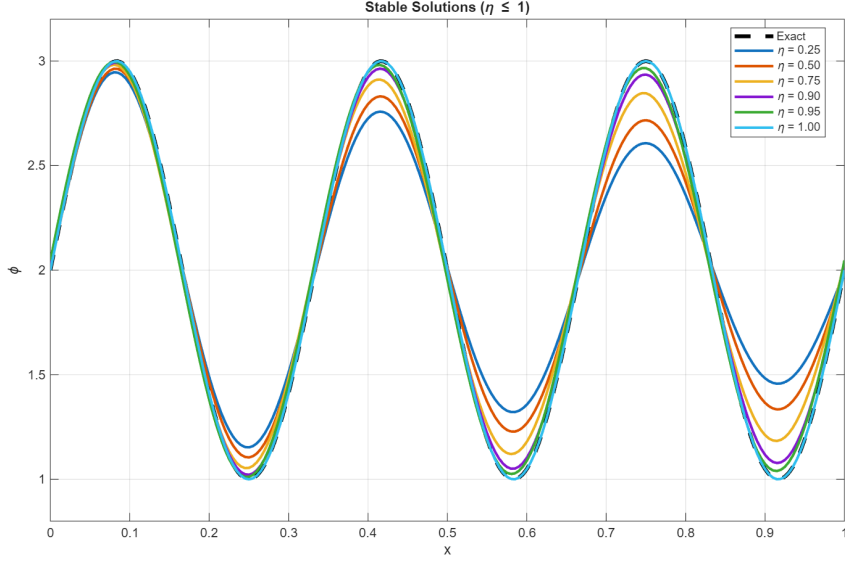


Figure 17: Stable numerical solutions for various values of the Courant number.

The figure clearly demonstrates the effect of numerical diffusion associated with first-order upwind discretisation. For small Courant numbers the wave amplitude decreases noticeably as the solution propagates through the domain. As η increases, the numerical solution approaches the exact travelling wave more closely. This behaviour is quantified by the amplitude ratio $\frac{A(t_{\text{end}})}{A(0)}$, which increases monotonically from approximately 0.897 for $\eta = 0.25$ to 1.000 for $\eta = 1.00$. The theoretical numerical diffusion coefficient of the explicit upwind scheme is

$$k_{\text{num}} = \frac{U\Delta x}{2}(1 - \eta). \quad (11)$$

This expression shows that the numerical diffusion decreases as η approaches unity. For $\eta = 1$ the artificial diffusion vanishes and the numerical solution reproduces the analytical travelling wave exactly. The simulation results confirm this prediction. At $\eta = 1$ the computed error is 4.61×10^{-16} , which is at the level of machine precision, and the amplitude ratio is exactly one. Therefore the optimal choice is

$$\boxed{\eta = 1} \quad (12)$$

since it satisfies the stability condition while simultaneously minimizing numerical damping.

3.3 Conclusions

The following conclusions can be drawn:

1. The simulations confirm the theoretical CFL stability requirement $\boxed{\eta \leq 1}$.
2. The solution becomes unstable for $\eta > 1$.
3. Numerical damping decreases as the Courant number increases.
4. The best performance is obtained for $\boxed{\eta = 1}$, for which the numerical solution is essentially identical to the analytical travelling-wave solution.

4 Exercise 4 – Unsteady diffusion and stability

Description

Consider the heat equation (pure diffusion, zero convection, $U = 0$)

$$\frac{\partial T}{\partial t} = \frac{\partial^2 T}{\partial x^2}, \quad 0 \leq x \leq 1, \quad 0 \leq t \leq 4,$$

with boundary conditions

$$T(0, t) = 0, \quad \frac{\partial T}{\partial x}(1, t) = 0$$

and initial condition

$$T(x, 0) = x.$$

This equation is solved with a finite-difference method on a grid with N grid points ($N = 10$ or 20). The Neumann condition is discretized with a mirror point. As time-integration method the generalized Crank–Nicolson method is used

$$\frac{T_j^{n+1} - T_j^n}{\delta t} = (1 - \omega) \frac{T_{j+1}^n - 2T_j^n + T_{j-1}^n}{h^2} + \omega \frac{T_{j+1}^{n+1} - 2T_j^{n+1} + T_{j-1}^{n+1}}{h^2}.$$

This gives for $\omega = 0$ the fully explicit method, for $\omega = 0.5$ the Crank–Nicolson method and for $\omega = 1$ the fully implicit method. **Required file:** `cfid_4.m` Typing the command `cfid_4` in Matlab asks for the input of the required parameters for this exercise: N , ω (omega) and δt . The quantity $T(1, t)$ is monitored.

Questions to be solved on the computer

(a)

First consider the fully explicit method ($\omega = 0$). Vary the number of grid points as $N = 10$ and $N = 20$. Solve the heat equation for various time steps, as indicated in the table below. Monitor whether (yes/no) the simulation is stable. In (d) you will be asked to explain the stability limit on the time step.

time step	0.001	0.00125	0.00126	0.0025	0.005	0.0051	0.01
$N = 10$ (stable?)	yes	yes	yes	yes	yes	no	no
$N = 20$ (stable?)	yes	yes	no	no	no	no	no

(b)

Next, use the Crank–Nicolson method ($\omega = 0.5$) and the generalized Crank–Nicolson method with $\omega = 0.6$. Use $N = 10$. Set the time step δt at 0.05 and 0.5; see the table below. For the larger time step the solution shows clear oscillations. Make a rough estimate for the damping factor (defined as the quotient of two successive amplitudes) of the oscillations.

Table 2: Crank–Nicolson and generalized Crank–Nicolson results.

ω	$\Delta t = 0.05$	$\Delta t = 0.5$	
	Stable?	Stable?	Damping Factor
0.5	yes	yes	0.9
0.6	yes	yes	0.65

(c)

Finally, again for $N = 10$, investigate the fully implicit method ($\omega = 1$). Use the same time steps as in (b). Observe whether the solution shows oscillations.

Table 3: Fully implicit method ($\omega = 1$).

ω	$\Delta t = 0.05$		$\Delta t = 0.5$	
	Stable?	Oscillations?	Stable?	Oscillations?
1.0	yes	no	yes	no

(d) Fourier Analysis of the Fully Explicit Method

For the fully explicit scheme we set

$$\omega = 0,$$

which yields

$$\frac{T_j^{n+1} - T_j^n}{\Delta t} = \frac{T_{j+1}^n - 2T_j^n + T_{j-1}^n}{h^2}.$$

Introducing the diffusion number

$$d = \frac{\Delta t}{h^2},$$

the scheme can be rewritten as

$$T_j^{n+1} = T_j^n + d(T_{j+1}^n - 2T_j^n + T_{j-1}^n).$$

Using a Fourier mode

$$T_j^n = \hat{T}^n e^{ij\theta},$$

substitution gives the amplification factor

$$g(\theta) = 1 + d(e^{i\theta} - 2 + e^{-i\theta}) = 1 - 2d(1 - \cos \theta).$$

For stability it is required that

$$|g(\theta)| \leq 1$$

for every wavenumber θ . The most restrictive case occurs when

$$\theta = \pi, \quad \cos \theta = -1.$$

Substitution yields

$$g(\pi) = 1 - 4d.$$

Hence

$$|1 - 4d| \leq 1.$$

This results in

$$0 \leq d \leq \frac{1}{2}.$$

Therefore the explicit method is stable if

$$\boxed{\Delta t \leq \frac{h^2}{2}}.$$

Maximum Stable Timestep

For a uniform grid the mesh spacing is

$$h = \frac{1}{N}.$$

For $N = 10$,

$$h = \frac{1}{10} = 0.1,$$

and therefore

$$\Delta t_{\max} = \frac{0.1^2}{2} = 0.005.$$

For $N = 20$,

$$h = \frac{1}{20} = 0.05,$$

and therefore

$$\Delta t_{\max} = \frac{0.05^2}{2} = 0.00125.$$

Thus,

$$\boxed{\Delta t_{\max}(N = 10) = 0.005}$$

and

$$\boxed{\Delta t_{\max}(N = 20) = 0.00125}.$$

Comparison with Numerical Results

The numerical results obtained in Question (a) are summarised in Table 4.

Table 4: Fully explicit method ($\omega = 0$).

time step	0.001	0.00125	0.00126	0.0025	0.005	0.0051	0.01
$N = 10$	yes	yes	yes	yes	yes	no	no
$N = 20$	yes	yes	no	no	no	no	no

The theoretical stability limits predict

$$\Delta t_{\max} = 0.005 \quad (N = 10),$$

and

$$\Delta t_{\max} = 0.00125 \quad (N = 20).$$

These values agree remarkably well with the numerical observations:

- For $N = 10$, simulations remain stable up to $\Delta t = 0.005$, while instability appears at $\Delta t = 0.0051$.
- For $N = 20$, simulations remain stable at $\Delta t = 0.00125$, while instability appears already at $\Delta t = 0.00126$.

Therefore the numerical experiments confirm the Fourier stability prediction with excellent accuracy.

Conclusion

The Fourier analysis yields the explicit stability condition

$$d \leq \frac{1}{2}$$

or equivalently

$$\Delta t \leq \frac{h^2}{2}.$$

For the two grids investigated this gives

$$\Delta t_{\max} = 0.005 \quad (N = 10)$$

and

$$\Delta t_{\max} = 0.00125 \quad (N = 20).$$

The stability limits observed in the numerical simulations coincide almost exactly with these theoretical values, demonstrating the validity of the Fourier stability analysis. The results also illustrate the strong dependence of the explicit timestep restriction on the mesh size: doubling the number of grid points reduces the maximum stable timestep by a factor of four.

(e) Fourier Stability Analysis of the Generalized Crank–Nicolson Method

The generalized Crank–Nicolson discretisation is

$$\frac{T_j^{n+1} - T_j^n}{\Delta t} = (1 - \omega) \frac{T_{j+1}^n - 2T_j^n + T_{j-1}^n}{h^2} + \omega \frac{T_{j+1}^{n+1} - 2T_j^{n+1} + T_{j-1}^{n+1}}{h^2}.$$

Introducing the diffusion number

$$d = \frac{\Delta t}{h^2},$$

and substituting the Fourier mode

$$T_j^n = \hat{T}^n e^{ij\theta},$$

gives the amplification factor

$$G = \frac{1 - 4(1 - \omega)d \sin^2(\theta/2)}{1 + 4\omega d \sin^2(\theta/2)}.$$

For stability it is required that

$$|G| \leq 1.$$

It can be shown that this condition is satisfied for all values of d whenever

$$\omega \geq \frac{1}{2}.$$

Therefore both

$$\omega = 0.5$$

(Crank–Nicolson) and

$$\omega = 0.6$$

are unconditionally stable. To understand the oscillations observed in the simulations, consider the limit

$$d \rightarrow \infty.$$

The amplification factor then approaches

$$G_\infty = -\frac{1 - \omega}{\omega}.$$

For the standard Crank–Nicolson scheme,

$$\omega = 0.5,$$

which gives

$$G_\infty = -1.$$

Hence high-frequency modes are not damped and simply change sign from one timestep to the next. This explains the pronounced oscillations observed for

$$\Delta t = 0.5.$$

For

$$\omega = 0.6,$$

the limiting amplification factor becomes

$$G_\infty = -\frac{0.4}{0.6} = -0.667.$$

The oscillatory modes are therefore damped and their amplitude decreases geometrically in time. The numerical simulations confirm this behaviour. For

$$\omega = 0.5$$

the solution remains stable but exhibits noticeable oscillations. A damping factor of approximately

$$\rho \approx 0.9$$

was observed. For

$$\omega = 0.6$$

the oscillations decay more rapidly and a damping factor of approximately

$$\rho \approx 0.65$$

was obtained. The resulting solution is visibly smoother and contains significantly smaller wiggles. The measured damping factors do not exactly match the theoretical amplification factors because Fourier analysis describes the evolution of a single Fourier mode, whereas the numerical solution consists of a superposition of modes. Furthermore, the amplitudes are measured from physical nodal values and are

influenced by the boundary conditions and the finite simulation time. Nevertheless, the qualitative agreement is excellent. Both the analysis and the simulations show that increasing ω beyond 0.5 introduces additional numerical damping while maintaining unconditional stability. Therefore:

$$\boxed{\omega \geq \frac{1}{2}}$$

guarantees unconditional stability, and increasing ω reduces oscillatory behaviour by damping the high-frequency modes more strongly.

(f) Wiggle-Free Condition

In part (e) the amplification factor of the generalized Crank–Nicolson method was found to be

$$G = \frac{1 - 4(1 - \omega)d \sin^2(\theta/2)}{1 + 4\omega d \sin^2(\theta/2)}.$$

Although the scheme is unconditionally stable for

$$\omega \geq \frac{1}{2},$$

stability alone does not guarantee a wiggle-free solution. Oscillations occur whenever the amplification factor becomes negative, since the sign of the corresponding Fourier mode changes between successive timesteps. To obtain a wiggle-free solution, it is therefore required that

$$G \geq 0$$

for all wave numbers. The denominator is always positive, so the condition is governed by the numerator,

$$1 - 4(1 - \omega)d \sin^2\left(\frac{\theta}{2}\right) \geq 0.$$

The most restrictive case corresponds to the highest-frequency mode,

$$\theta = \pi, \quad \sin^2\left(\frac{\theta}{2}\right) = 1.$$

Hence

$$1 - 4(1 - \omega)d \geq 0.$$

Rearranging gives

$$\boxed{d \leq \frac{1}{4(1 - \omega)}}.$$

This is the wiggle-free condition for the generalized Crank–Nicolson method. Writing the condition in terms of ω ,

$$\boxed{\omega \geq 1 - \frac{1}{4d}}.$$

For sufficiently large diffusion numbers, a larger value of ω is required to suppress oscillations. The result explains the observations of part (b). For

$$\omega = 0.5,$$

the numerator may become negative, producing oscillations. Although the scheme is stable, the solution exhibits visible wiggles because high-frequency modes are only weakly damped. For

$$\omega = 0.6,$$

the numerator remains positive over a larger range of diffusion numbers and the oscillations are considerably reduced. This was observed numerically through the smaller amplitude and larger damping of the oscillatory behaviour. As a special case, consider the fully implicit method

$$\omega = 1.$$

Substituting into the amplification factor gives

$$G = \frac{1}{1 + 4d \sin^2(\theta/2)}.$$

Since

$$1 + 4d \sin^2(\theta/2) > 0,$$

it follows that

$$0 < G \leq 1$$

for all Fourier modes. Therefore the amplification factor is always positive and no sign changes can occur between successive timesteps. Hence

$$\boxed{\omega = 1}$$

is inherently wiggle-free. This agrees exactly with the numerical results obtained in part (c), where the fully implicit method remained stable and no oscillations were observed for either

$$\Delta t = 0.05$$

or

$$\Delta t = 0.5.$$

In summary:

- Stability requires

$$\omega \geq \frac{1}{2}.$$

- A wiggle-free solution requires

$$d \leq \frac{1}{4(1 - \omega)}.$$

- Increasing ω increases numerical damping and reduces oscillations.
- The fully implicit method ($\omega = 1$) is both unconditionally stable and wiggle-free because all amplification factors remain positive.

Bonus – Stability of the Unsteady Convection–Diffusion Equation

Consider

$$\frac{\partial \phi}{\partial t} + U \frac{\partial \phi}{\partial x} = k \frac{\partial^2 \phi}{\partial x^2}, \quad 0 \leq x \leq L,$$

with Dirichlet boundary conditions

$$\phi(0, t) = \phi_0, \quad \phi(L, t) = \phi_1.$$

The spatial derivatives are discretised using central differences and the time derivative is discretised using the generalized Crank–Nicolson method.

Spatial Discretisation

Using a uniform grid spacing

$$h = \Delta x,$$

the central-difference approximation gives

$$\frac{\partial \phi}{\partial x} \approx \frac{\phi_{j+1} - \phi_{j-1}}{2h},$$

and

$$\frac{\partial^2 \phi}{\partial x^2} \approx \frac{\phi_{j+1} - 2\phi_j + \phi_{j-1}}{h^2}.$$

Introducing

$$\eta = \frac{U \Delta t}{h}, \quad d = \frac{k \Delta t}{h^2},$$

the semi-discrete operator becomes

$$L_h \phi_j = -\frac{\eta}{2}(\phi_{j+1} - \phi_{j-1}) + d(\phi_{j+1} - 2\phi_j + \phi_{j-1}).$$

Generalized Crank–Nicolson Discretisation

The generalized Crank–Nicolson method is

$$\phi^{n+1} = \phi^n + \Delta t \left[(1 - \omega) L_h \phi^n + \omega L_h \phi^{n+1} \right].$$

Rearranging gives

$$(I - \omega \Delta t L_h) \phi^{n+1} = (I + (1 - \omega) \Delta t L_h) \phi^n.$$

Fourier Analysis

Assume a Fourier mode

$$\phi_j^n = \hat{\phi}^n e^{ij\theta}.$$

The convective term becomes

$$-\frac{\eta}{2} (e^{i\theta} - e^{-i\theta}) = -i\eta \sin \theta.$$

The diffusive term becomes

$$d (e^{i\theta} - 2 + e^{-i\theta}) = -4d \sin^2 \left(\frac{\theta}{2} \right).$$

Therefore the spatial eigenvalue is

$$\lambda(\theta) = -i\eta \sin \theta - 4d \sin^2 \left(\frac{\theta}{2} \right).$$

The amplification factor becomes

$$G(\theta) = \frac{1 + (1 - \omega)\lambda(\theta)}{1 - \omega\lambda(\theta)}.$$

Substituting $\lambda(\theta)$,

$$G(\theta) = \frac{1 - (1 - \omega) [4d \sin^2 (\frac{\theta}{2}) + i\eta \sin \theta]}{1 + \omega [4d \sin^2 (\frac{\theta}{2}) + i\eta \sin \theta]}.$$

Special Cases

Explicit Euler ($\omega = 0$) The amplification factor reduces to

$$G = 1 - 4d \sin^2 \left(\frac{\theta}{2} \right) - i\eta \sin \theta.$$

Requiring

$$|G| \leq 1$$

for all wave numbers yields

$$\eta^2 \leq 2d$$

together with

$$d \leq \frac{1}{2}.$$

These are the classical Fourier stability conditions for the explicit central scheme.

Crank–Nicolson ($\omega = 0.5$) The amplification factor becomes

$$G = \frac{1 - \frac{1}{2}\lambda}{1 + \frac{1}{2}\lambda}.$$

For diffusion-dominated problems,

$$\Re(\lambda) \leq 0,$$

and one obtains

$$|G| \leq 1.$$

Therefore the Crank–Nicolson scheme is unconditionally stable. However,

$$G \rightarrow -1 \quad (d \rightarrow \infty),$$

so high-frequency modes remain weakly damped and oscillations may occur.

Generalized Crank–Nicolson ($\omega > 0.5$) For

$$\omega > \frac{1}{2},$$

the scheme remains unconditionally stable and additionally provides damping of high-frequency modes. The asymptotic limit is

$$G_{\infty} = -\frac{1-\omega}{\omega}.$$

Thus increasing ω introduces numerical damping.

Implicit Euler ($\omega = 1$) The amplification factor becomes

$$G = \frac{1}{1-\lambda}.$$

Since

$$\Re(\lambda) \leq 0,$$

it follows that

$$|G| \leq 1$$

for every timestep. Therefore

$$\omega = 1$$

is unconditionally stable and strongly damped.

Summary

For the unsteady convection–diffusion equation using central differencing in space and generalized Crank–Nicolson in time:

$$G = \frac{1 + (1-\omega)\lambda}{1 - \omega\lambda}$$

with

$$\lambda = -i\eta \sin \theta - 4d \sin^2 \left(\frac{\theta}{2} \right).$$

The resulting stability conditions are:

$$\eta^2 \leq 2d, \quad d \leq \frac{1}{2}$$

for the explicit scheme ($\omega = 0$), while

$$\omega \geq \frac{1}{2}$$

gives unconditional stability for the generalized Crank–Nicolson family. Furthermore,

$$\omega = 0.5$$

is non-dissipative and may exhibit oscillations, whereas

$$\omega > 0.5$$

introduces numerical damping, and

$$\omega = 1$$

produces the strongest damping and wiggle-free behaviour.